



Stomach & Duodenum — MCQ Practice

Physiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. An endocrinologist wants to locate the cells that synthesise gastrin in the stomach. Which cells are they?

EASY

- A. Parietal cells of the gastric glands
- B. G cells of the antrum
- C. D cells
- D. Mucous neck cells
- E. Chief cells

2. A biopsy shows loss of the cells that secrete pepsinogen. Which cell type is missing?

EASY

- A. Parietal cells
- B. G cells
- C. Chief cells
- D. Mucous neck cells
- E. ECL cells

3. A physiologist describes a hormone as the "master off switch" of the gut that inhibits most GI hormones and secretions. Which hormone is this?

EASY

- A. Secretin
- B. Gastrin
- C. Gastrin-releasing peptide
- D. Histamine
- E. Somatostatin

4. Approximately what share of gastric acid production occurs during the gastric phase?

EASY

- A. About 10%
- B. 60–70%
- C. 20–30%
- D. 40–50%
- E. More than 90%

5. A surgeon asks which part of the stomach is mainly responsible for gastric emptying into the duodenum. Which region is it?

MEDIUM

- A. The fundus, which has a thin wall and relaxes to store food
- B. The duodenum, which contracts to draw chyme out of the stomach
- C. The proximal stomach, through receptive relaxation
- D. The distal stomach (pylorus), which has a thick wall and strong, frequent peristalsis
- E. The body, which contracts weakly and infrequently

6. How do peristaltic waves in the distal stomach differ from mixing waves?

MEDIUM

- A. They move in the opposite direction and are weaker
- B. They occur only in the fundus
- C. They begin in the duodenum and travel backward
- D. They are identical but occur only after the stomach empties
- E. They move in the same direction but are stronger

7. A volunteer's stomach is distended with a nutrient-free balloon, without any digestion products. Which acid response is expected?

MEDIUM

- A. Acid production can still rise, because distension alone can stimulate it
- B. Only the cephalic phase responds, through the vagus
- C. Acid falls, because distension inhibits parietal cells
- D. No change, because acid needs digestion products to be released
- E. The intestinal phase is triggered, because the balloon acts as chyme

8. Which phase of acid secretion is initiated by chyme entering the small bowel and is described as poorly understood?

MEDIUM

- A. Basal phase
- B. Gastric phase
- C. Intestinal phase
- D. Cephalic phase
- E. Vagal phase

9. A drug prevents ECL cells from releasing histamine. What is the expected effect on acid secretion?

MEDIUM

- A. It rises, because histamine normally inhibits parietal cells
- B. It falls only during the cephalic phase
- C. It rises, because somatostatin release is abolished
- D. It falls, because histamine stimulates parietal cells and interacts with gastrin
- E. It is unchanged, because gastrin acts on chief cells only

10. A patient's gastric pH falls below 3.0. What is the effect on gastrin release?

MEDIUM

- A. Gastrin release rises only if alcohol is present
- B. Gastrin release is unchanged, since only vagal tone affects it
- C. Gastrin release is inhibited
- D. Gastrin release is stimulated to raise acid further
- E. Gastrin release is stimulated through falling pH acting on GRP

11. Acid and fat enter the duodenum. Which hormonal response follows from the lecture?

MEDIUM

- A. GRP release, which stimulates all GI secretions
- B. Suppression of somatostatin, so acid secretion rises
- C. Histamine release from mast cells, which stimulates parietal cells
- D. Gastrin release, which stimulates further acid secretion
- E. Somatostatin release, which inhibits acid secretion and other GI hormones

12. A patient with long-standing lack of gastric acid develops iron and calcium deficiency. Which function of HCl explains this?

MEDIUM

- A. HCl converts insoluble mucus into soluble mucus
- B. HCl activates intrinsic factor
- C. HCl stimulates gastrin secretion directly
- D. HCl helps Fe^{2+} and Ca^{2+} absorption
- E. HCl is required for pancreatic lipase to act

13. Why is the duodenum often bypassed in gastric bypass surgery for obesity?

MEDIUM

- A. It stores food, so bypassing it reduces meal volume
- B. It absorbs nutrients even more than the stomach, so bypassing it decreases absorption
- C. It makes intrinsic factor, so bypassing it prevents fat absorption
- D. It secretes gastrin, so bypassing it reduces acid
- E. It is the main site of mechanical breakdown of food

14. A researcher analyses small-intestinal secretion (1-3 L/day). Which set lists functions given in the lecture?

MEDIUM

- A. Protection by mucus, digestion by enzymes, and dilution
- B. Absorption of bile salts, secretion of secretin and mixing of chyme
- C. Production of intrinsic factor, gastrin release and receptive relaxation
- D. Sterilisation by HCl, neutralisation of acid and storage of chyme
- E. Activation of pepsinogen, emulsification of fat and vitamin B12 absorption

15. A patient has lost the mucous neck cells of the gastric glands. Which consequence is most directly expected?

HARD

- A. Loss of acid secretion, because mucous neck cells secrete HCl
- B. Loss of protein digestion, because they secrete pepsinogen
- C. Loss of vitamin B12 absorption, because they secrete intrinsic factor
- D. Loss of the mucus barrier and bicarbonate buffering, exposing the epithelium to acid
- E. Excess gastrin release, because they secrete somatostatin

16. A lesion stops the vagus from releasing ACh onto the stomach. Which change is expected?

HARD

- A. Loss of the intestinal phase, because it is entirely vagal
- B. Loss of the cephalic contribution (20–30% of acid) and reduced vagally stimulated GRP release
- C. Increased acid, because somatostatin is no longer released
- D. No change in total acid, because distension alone replaces all phases
- E. Loss of the gastric phase (60–70% of acid) only

17. In an experiment, histamine release from ECL cells is blocked and gastrin is then given. Which result best fits the lecture?

HARD

- A. Acid rises normally, because histamine only affects chief cells
- B. Acid falls below baseline, because gastrin inhibits parietal cells
- C. Acid still rises, though less than usual, because gastrin also acts directly on parietal cells
- D. Acid rises only via the vagus, because gastrin has no gastric targets
- E. No acid response at all, because gastrin acts only through histamine

18. A patient with a high-output intestinal fistula is given octreotide, which the lecture lists under somatostatin. Which somatostatin action best explains the reduced output?

HARD

- A. Inhibition of GI and pancreatic secretions and of intestinal motility
- B. Stimulation of gastrin release from G cells
- C. Stimulation of all GIT secretions and motility
- D. Stimulation of GI mucosal growth
- E. Potentiation of histamine on parietal cells

19. Which description of gastric mechanics is correct?

HARD

- A. The proximal stomach stores food by receptive relaxation; each peristaltic wave passes a few mL of mostly fluid chyme into the duodenum while most is forced back for further mixing
- B. The distal stomach stores food by receptive relaxation, and each wave forces the solid centre of the chyme through the pylorus first
- C. The proximal stomach mixes food with strong, frequent peristalsis, and each wave empties most of the chyme into the duodenum at once
- D. The distal stomach holds large volumes while the proximal stomach empties chyme into the duodenum with each contraction
- E. The whole stomach contracts weakly and infrequently, so gastric emptying relies mainly on gravity and receptive relaxation

20. After a fatty, acidic meal, chyme with fat and acid enters the duodenum. Which change in gastric acid secretion is expected, and why?

HARD

- A. It rises, because CCK and secretin stimulate the G cells
- B. It rises, because acid in the duodenum releases gastrin directly
- C. It is unchanged, because duodenal signals affect only pancreatic secretion
- D. It falls, because duodenal acid stimulates GRP release from the antrum
- E. It falls, because fat and acid in the duodenum release somatostatin, and secretin and CCK inhibit gastrin release

✓ Answer key

Q	Answer	Level	Why (lecture slide)
1	B	EASY	Gastrin is synthesised by G cells in the antrum. [Slides 8, 11]
2	C	EASY	Chief cells secrete pepsin(ogen) and gastric lipase. [Slide 8]
3	E	EASY	Somatostatin is the "master off switch": inhibits most GI hormones, GI and pancreatic secretion, and intestinal motility. [Slide 14]
4	B	EASY	Gastric phase accounts for 60–70% of acid production (cephalic 20–30%, intestinal ~10%). [Slides 19–21]
5	D	MEDIUM	The distal area has a thick wall with strong frequent peristalsis and is responsible for gastric emptying. [Slide 5]
6	E	MEDIUM	Peristaltic waves move in the same direction and in the same way as mixing waves, but are stronger. [Slide 6]
7	A	MEDIUM	Gastric phase begins with distension; distension alone can increase acid production. [Slide 20]
8	C	MEDIUM	The intestinal phase is poorly understood, starts when chyme enters the small bowel and accounts for ~10% of acid. [Slide 21]
9	D	MEDIUM	Histamine (ECL cells) stimulates parietal cells; gastrin has a potentiating interaction with histamine. [Slides 12, 16]
10	C	MEDIUM	pH < 3.0 inhibits gastrin release; pH > 3.0 stimulates it. [Slide 11]
11	E	MEDIUM	Fats, protein and acid in the duodenum release somatostatin, the "master off switch". [Slides 13, 14]
12	D	MEDIUM	One of the six functions of gastric HCl: helps Fe ²⁺ and Ca ²⁺ absorption. [Slide 23]
13	B	MEDIUM	The duodenum absorbs nutrients even more than the stomach, so it is frequently bypassed to decrease absorption. [Slide 25]
14	A	MEDIUM	Small-intestinal secretion: protective effect by mucus, digestion by enzymes (peptidase, sucrase, lipase), dilution. [Slide 26]
15	D	HARD	Mucous neck cells secrete mucus (physical barrier) and bicarbonate (buffers acid to prevent epithelial damage). [Slide 8]
16	B	HARD	The cephalic phase acts via vagal ACh on parietal cells (20–30%); GRP release is stimulated by vagal stimulation. [Slides 15, 19]
17	C	HARD	Gastrin acts directly on parietal cells and also has a potentiating interaction with histamine (the "less than usual" is an inference). [Slide 12]
18	A	HARD	Somatostatin inhibits pancreatic and GI secretions and intestinal motility; octreotide decreases fistula output. [Slide 14]
19	A	HARD	Proximal: thin wall, receptive relaxation. Distal: strong peristalsis; a few mL of mostly fluid chyme pass, most is forced back. [Slides 5, 6]
20	E	HARD	Fats, protein and acid in the duodenum release somatostatin; secretin and CCK (duodenal hormones) are listed as inhibitors of gastrin. [Slides 11, 13, 25]